

Def. A Homotopy of paths in X is a family

$$f_t: I \rightarrow X, \quad 0 \leq t \leq 1 \quad \text{such that}$$

- 1) For all $0 \leq t \leq 1$, $f_t(0) = x_0$ and $f_t(1) = x_1$.
- 2) $F: I \times I \rightarrow X; (s, t) \mapsto f_t(s)$ is continuous.

Given two paths P and g from $x_0 \in X$ to x_1 , they are said to be homotopic if there exists a homotopy of paths f_t satisfying $P = f_0$ and $g = f_1$.

denoted as $P \simeq g$.

Example. Linear Homotopy

Given two paths P and g in $\mathbb{R}^n$ having the same endpoints x_0 and x_1 , define

$$f_t: I \rightarrow \mathbb{R}^n; s \mapsto (1-t)P(s) + t g(s)$$

for each $0 \leq t \leq 1$. Then, given t ,

$$f_t(0) = (1-t)P(0) + t g(0) = x_0 \quad \text{and} \quad f_t(1) = (1-t)P(1) + t g(1) = x_1$$

(In actually, for any $s \in [0, 1]$, $P(s) = g(s) \Rightarrow f_t(s) = P(s)$)

And, to show $F: I \times I \rightarrow \mathbb{R}^n; (s, t) \mapsto f_t(s)$ is continuous, let $M > 0$ be a bound for P and g .

Let $(s_0, t_0) \in I \times I$ be given. Given $\epsilon > 0$, there is $\delta > 0$ such that

$$\|s - s_0\| < \delta \Rightarrow \|f_{t_0}(s) - f_{t_0}(s_0)\| < \epsilon \quad \text{and}$$

$$\begin{aligned} \|t - t_0\| < \delta \Rightarrow \|f_t(s) - f_{t_0}(s)\| &= \|(1-t)P(s) + t g(s) - [(1-t_0)P(s) + t_0 g(s)]\| \\ &= \|(t_0 - t)P(s) + (t - t_0)g(s)\| \\ &\leq |t_0 - t| \cdot (\|P(s)\| + \|g(s)\|) < \epsilon. \end{aligned}$$

Now,

$$\|(s, t) - (s_0, t_0)\| < \delta \Rightarrow \|f_t(s) - f_{t_0}(s_0)\| \leq \|f_t(s) - f_{t_0}(s)\| + \|f_{t_0}(s) - f_{t_0}(s_0)\| < 2\epsilon.$$

So $P \simeq g$ $\square$

Prop. Let X be a Topological Space. Given two points x_0 and x_1 in X , define an relation on the set of paths from x_0 to x_1 as: $p \sim q \Leftrightarrow p \simeq q$. Then, this relation is equivalence relation.

Proof. Reflexivity is evident, by $f_t = f$, the constant homotopy.

Symmetry is given by the inverse homotopy: if $f_0 \simeq f_1$ via f_t , then $f_1 \simeq f_0$ via f_{1-t} .

To show the Transitivity, suppose that $f_0 \simeq f_1$ via f_t and $g_0 \simeq g_1$ via g_t , with $f_1 = g_0$.

Define $h: I \times I \rightarrow X: (s, t) \mapsto \begin{cases} f_{2t}(s) & \text{if } 0 \leq t \leq \frac{1}{2} \\ g_{2t-1}(s) & \text{if } \frac{1}{2} \leq t \leq 1. \end{cases}$

It is well-defined since $f_1 = g_0$. And, Continuity is given by the Pasting Lemma.

Def. A path $p: I \rightarrow X$ is called a loop if $p(0) = p(1) = x_0 \in X$, and the point x_0 is called the basepoint.

The collection of all homotopy classes $[f]$ of loops $f: I \rightarrow X$ at the basepoint x_0 ,

$$\pi_1(X, x_0) \stackrel{\text{def}}{=} \{ [f] \mid f \text{ is a loop at } x_0 \}$$

is called the fundamental group under the operation

$$[f][g] \stackrel{\text{def}}{=} [f \cdot g] \text{ where } f \cdot g: I \rightarrow X: s \mapsto \begin{cases} f(2s) & \text{if } 0 \leq s \leq \frac{1}{2} \\ g(2s-1) & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases}$$

This $\pi_1(X, x_0)$ is well-defined as a set clearly, and the operation is also well defined by: Suppose that $[f_1] = [f_2]$ and $[g_1] = [g_2]$. Then $f_1 \simeq f_2$ and $g_1 \simeq g_2$ via $\hat{f}_t$ and $\hat{g}_t$.

$$\begin{aligned} [f_1][g_1] &= [f_1 \cdot g_1] = [f_2 \cdot g_2] \text{ because } f_1 \cdot g_1 \simeq f_2 \cdot g_2 \text{ via } \hat{f}_t \cdot \hat{g}_t \\ &= [f_2][g_2] \end{aligned}$$

Moreover, the operation satisfies the Group Axiom, the proof lies in the next Page.

proof of the fundamental group.

First, given a path $f: I \rightarrow X$, define a reparametrization of a path f as a composition $f \circ \varphi$ where $\varphi: I \rightarrow I$ is any continuous map s.t. $\varphi(0)=0$ and $\varphi(1)=1$.

The reparametrization preserves homotopy, that is,

$f \circ \varphi \simeq f$ via $f \circ \varphi_t$ where $\varphi_t(s) = (1-t)\varphi(s) + ts$, so that $\varphi_0 = \varphi$ and $\varphi_1 = \text{id}$.

Associativity. Let f, g, h be paths with $f(1) = g(0)$ and $g(1) = h(0)$.

Then, the composition $(f \cdot g) \cdot h = f \cdot (g \cdot h)$ are defined. Moreover,

$f \cdot (g \cdot h)$ is a reparametrization of $(f \cdot g) \cdot h$, because

$$f \cdot (g \cdot h)(s) = \begin{cases} f(2s) & \text{if } 0 \leq s \leq \frac{1}{2} \\ g(4s-2) & \text{if } \frac{1}{2} \leq s \leq \frac{3}{4} \\ h(4s-3) & \text{if } \frac{3}{4} \leq s \leq 1 \end{cases} \quad \text{and}$$

$$(f \cdot g) \cdot h(s) = \begin{cases} f(4s) & \text{if } 0 \leq s \leq \frac{1}{4} \\ g(4s-1) & \text{if } \frac{1}{4} \leq s \leq \frac{1}{2} \\ h(2s-1) & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases} \quad \text{therefore}$$

$$f \cdot (g \cdot h) = (f \cdot g) \cdot h \circ \varphi \quad \text{where} \quad \varphi(s) = \begin{cases} \frac{1}{2}s & \text{if } 0 \leq s \leq \frac{1}{2} \\ s - \frac{1}{4} & \text{if } \frac{1}{2} \leq s \leq \frac{3}{4} \\ 2s-1 & \text{if } \frac{3}{4} \leq s \leq 1 \end{cases}$$

so $f \cdot (g \cdot h) \simeq (f \cdot g) \cdot h$, the associative is proven

Identity. Given a path $f: I \rightarrow X$, let C be the constant path, as

$$C_i: I \rightarrow X: s \mapsto f(i) \quad (i=0,1)$$

Then $f \cdot C_1$ is a reparametrization of f via

$$\varphi: I \rightarrow I: s \mapsto \begin{cases} 2s & \text{if } 0 \leq s \leq \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases}$$

and similarly $C_0 \cdot f$ is also a reparametrization, so, $C_0 \cdot f \simeq f \cdot C_1 \simeq f$.

For a loop, the endpoints are fixed, so $C(s) = x_0$ is two-sided identity in $\pi_1(X, x_0)$.

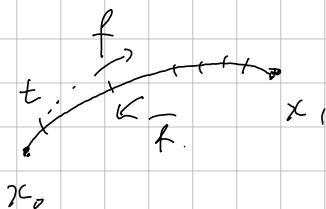
Inverse. Given a path f from x_0 to x_1 , define the inverse path

$$\bar{f}(s) = f(1-s) \quad \text{so that } \bar{f}(0) = f(1), \bar{f}(1) = f(0)$$

Then $f \cdot \bar{f}$ is homotopic to a constant path $C(s) = f(0)$, via $h_t = f_t \cdot \bar{f}_t$ where

$$f_t: I \rightarrow X: s \mapsto \begin{cases} f(s) & \text{if } 0 \leq s \leq 1-t \\ f(1-t) & \text{if } 1-t \leq s \leq 1 \end{cases}$$

similarly, for a loop, $f \cdot \bar{f} \simeq \bar{f} \cdot f \simeq C(\equiv x_0)$.



[Change of basePoint.

[Def. Let X be a space, and $x_0, x_1 \in X$ be points contained a same path-component.

Let $h: I \rightarrow X$ be a path from x_0 to x_1 , with the inverse curve $\bar{h}(s) = h(1-s)$.

Then, for any loop at x_1 , the composition $h \cdot f \cdot \bar{h}$ is a loop at x_0 .

Rigorously, $(h \cdot f) \cdot \bar{h} \neq h \cdot (f \cdot \bar{h})$ but $(h \cdot f) \cdot \bar{h} \simeq h \cdot (f \cdot \bar{h})$. Therefore, the map

$$\beta_h: \pi_1(X, x_1) \rightarrow \pi_1(X, x_0): [f] \mapsto [h \cdot f \cdot \bar{h}]$$

is well-defined, and is called a change of basepoint.

Moreover, for path-connected x_0 and x_1 , the map β_h is an Isomorphism.

Homomorphism: $\beta_h[f \cdot g] = [h \cdot f \cdot g \cdot \bar{h}] = [h \cdot f \cdot \bar{h} \cdot h \cdot g \cdot \bar{h}] = \beta_h[f] \beta_h[g]$.

Invertible: β_h satisfies

$$\beta_h \beta_{\bar{h}}[f] = \beta_h[\bar{h} \cdot f \cdot h] = [h \cdot \bar{h} \cdot f \cdot h \cdot \bar{h}] = [f].$$

and similarly $\beta_{\bar{h}} \beta_h[f] = [f]$.

[Def. A space X is called 'if it is path-connected and has trivial fundamental group.

[Def. Given a space X , a covering space of X consists of a space $\tilde{X}$ and a map $p: \tilde{X} \rightarrow X$ s.t.

For each $x \in X$, there is an open neighborhood U of x in X satisfying

$$p^{-1}[U] = \bigsqcup_{i \in I} \mathcal{O}_i, \text{ each } \mathcal{O}_i \text{ s.t. } \mathcal{O}_i \stackrel{\text{Homeo}}{\cong} U \text{ via } p.$$

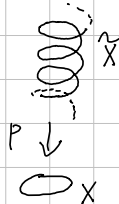
[Example. Given a space S^1 , let $p: \mathbb{R} \rightarrow S^1: s \mapsto (\cos 2\pi s, \sin 2\pi s)$.

Let $(1, 0) \in S^1$ be given. Then, for a neighborhood $B_1(1, 0) \cap S^1$,

$$p^{-1}[B_1(1, 0) \cap S^1] = \bigsqcup_{n \in \mathbb{Z}} (n + (-\frac{1}{8}, \frac{1}{8}))$$

and for each $n \in \mathbb{Z}$,

$$(n + (-\frac{1}{8}, \frac{1}{8})) \cong B_1(1, 0) \cap S^1 \text{ via } p.$$



Prop. Let X and Y are Path-Connected. Then,

$$\pi_1(X \times Y) \cong \pi_1(X) \times \pi_1(Y)$$

Proof. Notice that: $f: Z \rightarrow X \times Y$ is continuous iff $p_1 \circ f$ and $p_2 \circ f$ both are continuous, where $p_1: X \times Y \rightarrow X$ and $p_2: X \times Y \rightarrow Y$ are Projections.

Therefore, there is a one-to-one Correspondence

$$\{f: I \rightarrow X \times Y \mid f \text{ is a loop at } (x_0, y_0)\}$$



$$\{g: I \rightarrow X \mid g \text{ is a loop at } x_0 \in X\} \times \{h: I \rightarrow Y \mid h \text{ is a loop at } y_0 \in Y\},$$

Under this Correspondence, we can define a bijective

$$\pi_1(X \times Y, (x_0, y_0)) \rightarrow \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

$$: [f] \mapsto ([g], [h]) \quad \text{where } f = (g, h)$$

is a bijective group Homomorphism, so proved

The Fundamental Group of the Circle.

Thm. $\pi_1(S^1)$ is an infinite cyclic group generated by $[w]$, where

$$w(s) = (\cos 2\pi s, \sin 2\pi s) \text{ based at } (1, 0).$$

Note that $[w]^n = [w_n]$ where $w_n(s) = (\cos 2\pi ns, \sin 2\pi ns)$ for each $n \in \mathbb{N}$.

Consider $p: \mathbb{R} \rightarrow S^1; s \mapsto (\cos 2\pi s, \sin 2\pi s)$ and $\tilde{w}_n: \mathbb{I} \rightarrow \mathbb{R}; s \mapsto ns$.

Then p is a covering map, and $\tilde{w}_n$ is called the lift of w_n , satisfying

$$w_n = p \circ \tilde{w}_n.$$

Proof. Let $f: \mathbb{I} \rightarrow S^1$ be a loop at the basepoint $x_0 = (1, 0)$.

Since S^1 is path-connected, $[f] \in \pi_1(S^1)$.

Thm. Let X be a space and let $w: I \rightarrow X$ be a path from x_0 to x_1 .
Then, a map $\hat{w}: \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$
 $[\alpha] \mapsto [\bar{w} * \alpha * w]$

is a Group Isomorphism.

Thm. Let $f: X \rightarrow Y$ be a continuous map such that $f(x_0) = y_0$.
Then, f induces a group homomorphism

$$f_{\#}: \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0); [\alpha] \mapsto [f \circ \alpha].$$

Proof. Notice that given a loop $\alpha: I \rightarrow X$ at x_0 , a composition $f \circ \alpha: I \rightarrow Y$ satisfies $(f \circ \alpha)(0) = (f \circ \alpha)(1) = f(x_0) = y_0$, so $f \circ \alpha$ is a loop at y_0 .

Let $[\alpha], [\beta] \in \pi_1(X, x_0)$. If $[\alpha] = [\beta]$, then there is a Homotopy

$F: I \times I \rightarrow X$ s.t. $F(x, 0) = \alpha(x)$ and $F(x, 1) = \beta(x)$, so that

$f \circ F: I \times I \rightarrow Y$ satisfies $(f \circ F)(x, 0) = (f \circ \alpha)(x)$ and $(f \circ F)(x, 1) = (f \circ \beta)(x)$

therefore, $\alpha \stackrel{F}{\sim} \beta$ implies $f \circ \alpha \stackrel{f \circ F}{\sim} f \circ \beta$, thus $[f \circ \alpha] = [f \circ \beta]$.

It proves the $f_{\#}$ is well-defined. Now,

$$\begin{aligned} f_{\#}([\alpha] * [\beta]) &= f_{\#}([\alpha * \beta]) = [f \circ (\alpha * \beta)] = [(f \circ \alpha) * (f \circ \beta)] = [f \circ \alpha] * [f \circ \beta] \\ &= f_{\#}([\alpha]) * f_{\#}([\beta]) \quad \square \end{aligned}$$

Thm. Let $f: X \rightarrow Y$ be a Homotopy equivalence such that $f(x_0) = y_0$.
Then, the induced map $f_{\#}$ is an isomorphism.